

Objective: A temperature controlled fan circuit.

Components: U741 op-amp, NPN transistor, potentiometer, Thermistor, voltage controlled fan, diode, resistors, source.

Theory: An op-amp is an electronic component which is used to amplify a small signal. When the voltage on the non-inverting is lower than the reference voltage in inverting point, the op-amp give low signal, in this case 0v. When voltage at non-inverting goes over the reference voltage, the signal goes through. The Reference voltage is set using a basic voltage divider where the latter resistor is thermistor. When temperature is increased, it's internal resistor gets low causing the reference voltage go lower. A potentiometer in the non-inverting point used to set the voltage going into the positive to control the temp to turn on the fan.

When the voltage gets amplified, it is passed through a resistor to generate current which goes to base of the BJT (acts as a switch). When base current is passed through current passed through emitter & collector. Then the fan turns on. If the voltage 0 from the amplifier, the switch is off, thus the circuit is open. So the fan stays off. A flyback diode is used protect the BJT. When the circuit is turned off after begin on, a voltage may go back to transistor from the motor of the fan which would cause the transistor to get wasted.

Circuit:

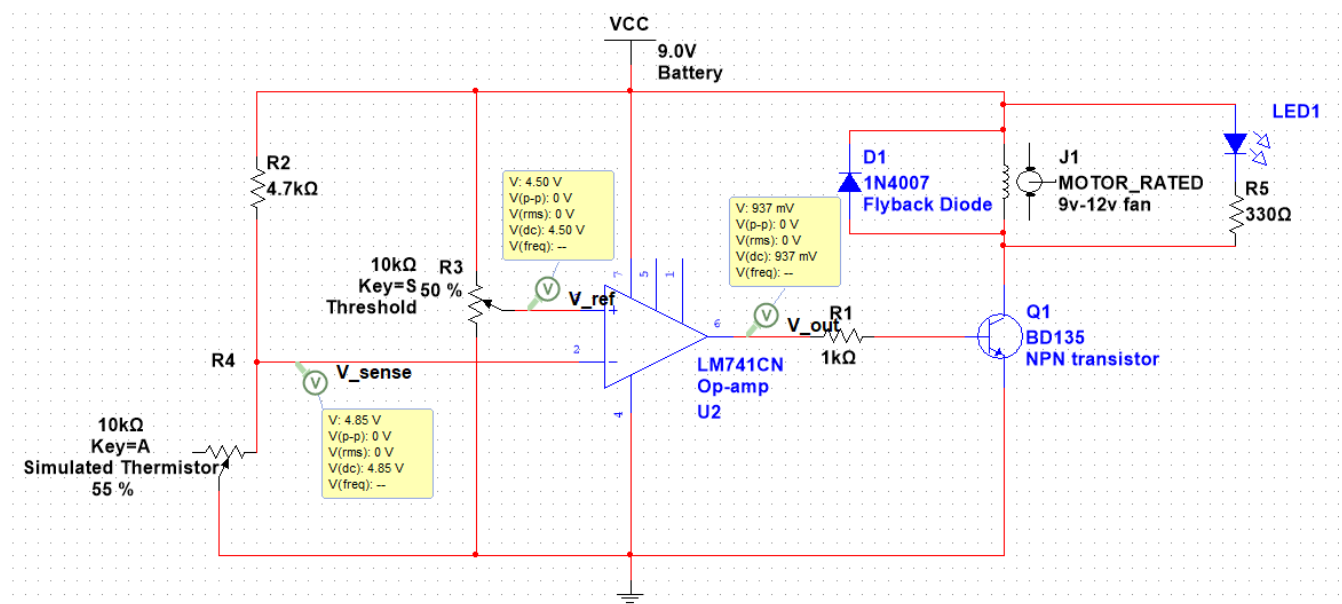


fig: voltage at non-inverting lower than Vref

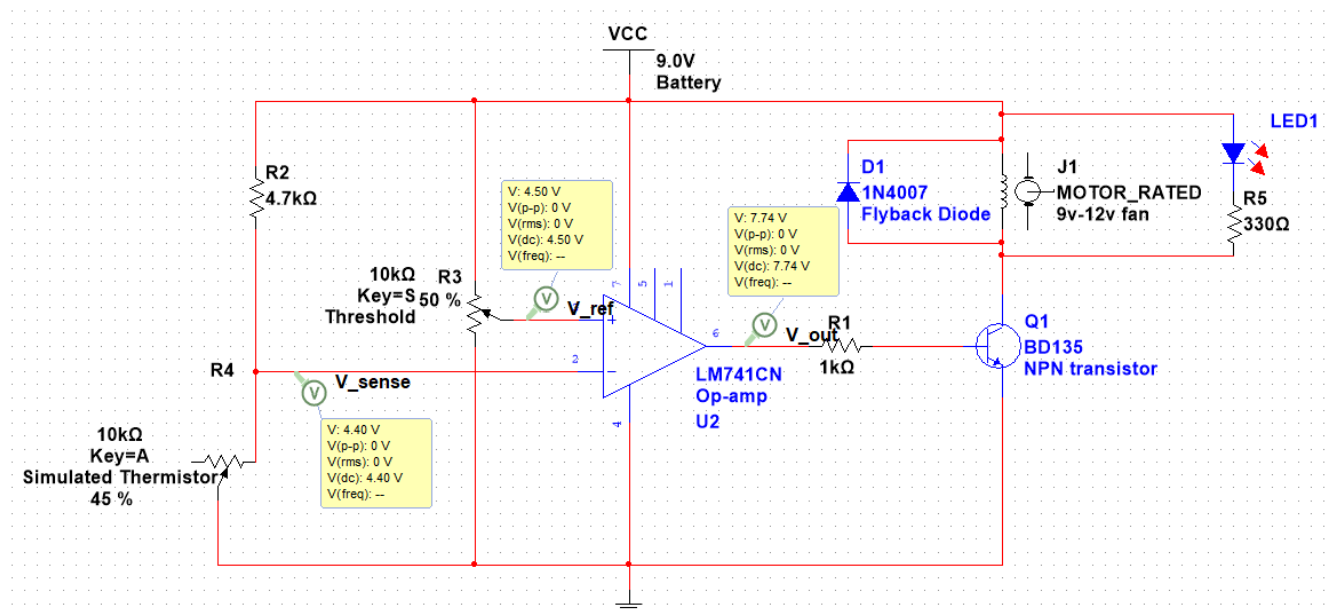


fig: voltage at non-inverting higher than V_{ref}